

Prompt Engineering

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What is the Prompt Engineering

- Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Prompt Elements

- **Instruction** a specific task or instruction you want the model to perform
- **Context** external information or additional context that can steer the model to better responses
- **Input Data** the input or question that we are interested to find a response for
- **Output Indicator** the type or format of the output.

Why it is Important?

- **Quality in, quality out** : Large-language models are like ultra-talented interns.
- **Cut hallucinations**: Adding context, constraints, or tool instructions drastically reduces made-up facts.
- **Leverage hidden capabilities**: Few-shot examples, role instructions, or ReAct loops can unlock specialized behaviors
- **Cost & latency control**: Tight prompts = shorter outputs = lower token bills and faster responses. In production chatbots, shaving 20–30 % tokens can be the difference between profitable and not.
- **Reusability & scaling**: A well-engineered prompt becomes a reusable component—that multiple teams or pipelines can plug in.

**Prompting is an iterative
process**

General Tips

Start Simple

The Instruction

Input

```
### Instruction ###  
Translate the text below to Spanish:  
Text: "hello!"
```

Output

```
¡Hola!
```


Be Specific

Input

Extract the name of places in the following text.

Desired format:

Place: <comma_separated_list_of_places>

Input: "Although these developments are encouraging to researchers, much is still a mystery. "We often have a black box between the brain and the effect we see in the periphery," says Henrique Veiga-Fernandes, a neuroimmunologist at the Champalimaud Centre for the Unknown in Lisbon. "If we want to use it in the therapeutic context, we actually need to understand the mechanism.""

Output

Place: Champalimaud Centre for the Unknown, Lisbon

Avoid Impreciseness

Bad Example ❌

Explain the concept prompt engineering. Keep the explanation short, only a few sentences, and don't be too descriptive.

Good Example ✅

Use 2–3 sentences to explain the concept of prompt engineering to a high school student.



To do or not to do?

The Answer is : To do

Input

The following is an agent that recommends movies to a customer. DO NOT ASK FOR INTERESTS. DO NOT ASK FOR PERSONAL INFORMATION.
Customer: Please recommend a movie based on my interests.
Agent:

Output

Sure, I can recommend a movie based on your interests. What kind of movie would you like to watch? I can recommend action, comedy, romance, or something else? 

Prompt Techniques

Zero-Shot Prompting

Prompt

```
Classify the text into neutral, negative or positive.  
Text: I think the vacation is okay.  
Sentiment:
```

Output

```
Neutral
```


Few-Shot Prompting

Prompt

```
This is awesome! // Negative  
This is bad! // Positive  
Wow that movie was rad! // Positive  
What a horrible show! //
```

Output

```
Negative
```

**Few-shot is Good but not for
reasoning Questions!**

Chain-of-thoughts

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Zero-shot CoT Prompting

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

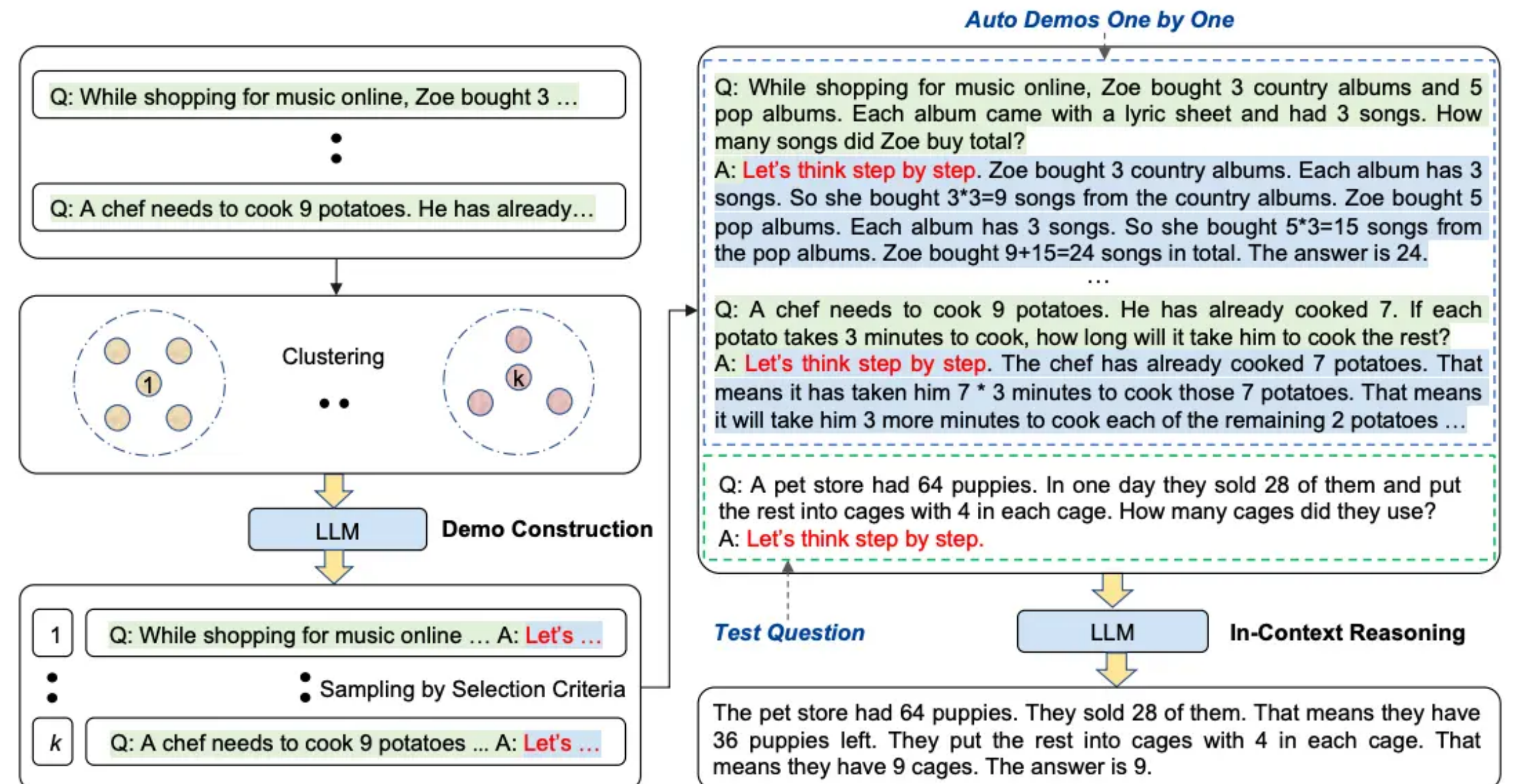
A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Auto-CoT

Auto-CoT consists of two main stages:

- Stage 1): Question clustering: partition questions of a given dataset into a few clusters
- Stage 2): Demonstration sampling: select a representative question from each cluster and generate its reasoning chain using Zero-Shot-CoT with simple heuristics



Meta Prompting

Meta Prompting is an advanced prompting technique that focuses on the structural and syntactical aspects of tasks and problems rather than their specific content details

This goal with meta prompting is to construct a more abstract, structured way of interacting with large language models (LLMs), emphasizing the form and pattern of information over traditional content-centric methods

Meta Prompting Example

The advantages of Meta Prompting over few-shot promoting include:

- 1. Token efficiency:** Reduces the number of tokens required by focusing on structure rather than detailed content.
- 2. Fair comparison:** Provides a more fair approach for comparing different problem-solving models by minimizing the influence of specific examples.
- 3. Zero-shot efficacy:** Can be viewed as a form of zero-shot prompting, where the influence of specific examples is minimized.

Problem Statement:

- **Problem:** [question to be answered]

Solution Structure:

1. Begin the response with "Let's think step by step."
2. Follow with the reasoning steps, ensuring the solution process is broken down clearly and logically.
3. End the solution with the final answer encapsulated in a LaTeX-formatted box, $\boxed{\dots}$, for clarity and emphasis.
4. Finally, state "The answer is [final answer to the problem].", with the final answer presented in LaTeX notation.

Figure 1: A structure meta prompt presented in markdown format for solving MATH [17] problems.

Problem: Find the domain of the expression $\frac{\sqrt{x-2}}{\sqrt{5-x}}$.

Solution: The expressions inside each square root must be non-negative. Therefore, $x - 2 \geq 0$, so $x \geq 2$, and $5 - x \geq 0$, so $x \leq 5$. Also, the denominator cannot be equal to zero, so $5 - x > 0$, which gives $x < 5$. Therefore, the domain of the expression is $\boxed{[2, 5)}$. Final Answer: The final answer is $[2, 5)$. I hope it is correct.

Problem: If $\det \mathbf{A} = 2$ and $\det \mathbf{B} = 12$, then find $\det(\mathbf{AB})$.

Solution: We have that $\det(\mathbf{AB}) = (\det \mathbf{A})(\det \mathbf{B}) = (2)(12) = \boxed{24}$. Final Answer: The final answer is 24. I hope it is correct.

...

Prompt Chaining

- Prompt chaining is useful to accomplish complex tasks which an LLM might struggle to address if prompted with a very detailed prompt. In prompt chaining, chain prompts perform transformations or additional processes on the generated responses before reaching a final desired state.
- Once those subtasks have been identified, the LLM is prompted with a subtask and then its response is used as input to another prompt.

Prompt Chaining Example

Prompt

```
You are a helpful assistant. Your task is to help answer a question given in a document. The first step is to extract quotes relevant to the question from the document, delimited by ####. Please output the list of quotes using <quotes></quotes>. Respond with "No relevant quotes found!" if no relevant quotes were found.
```

```
####
```

```
{{document}}
```

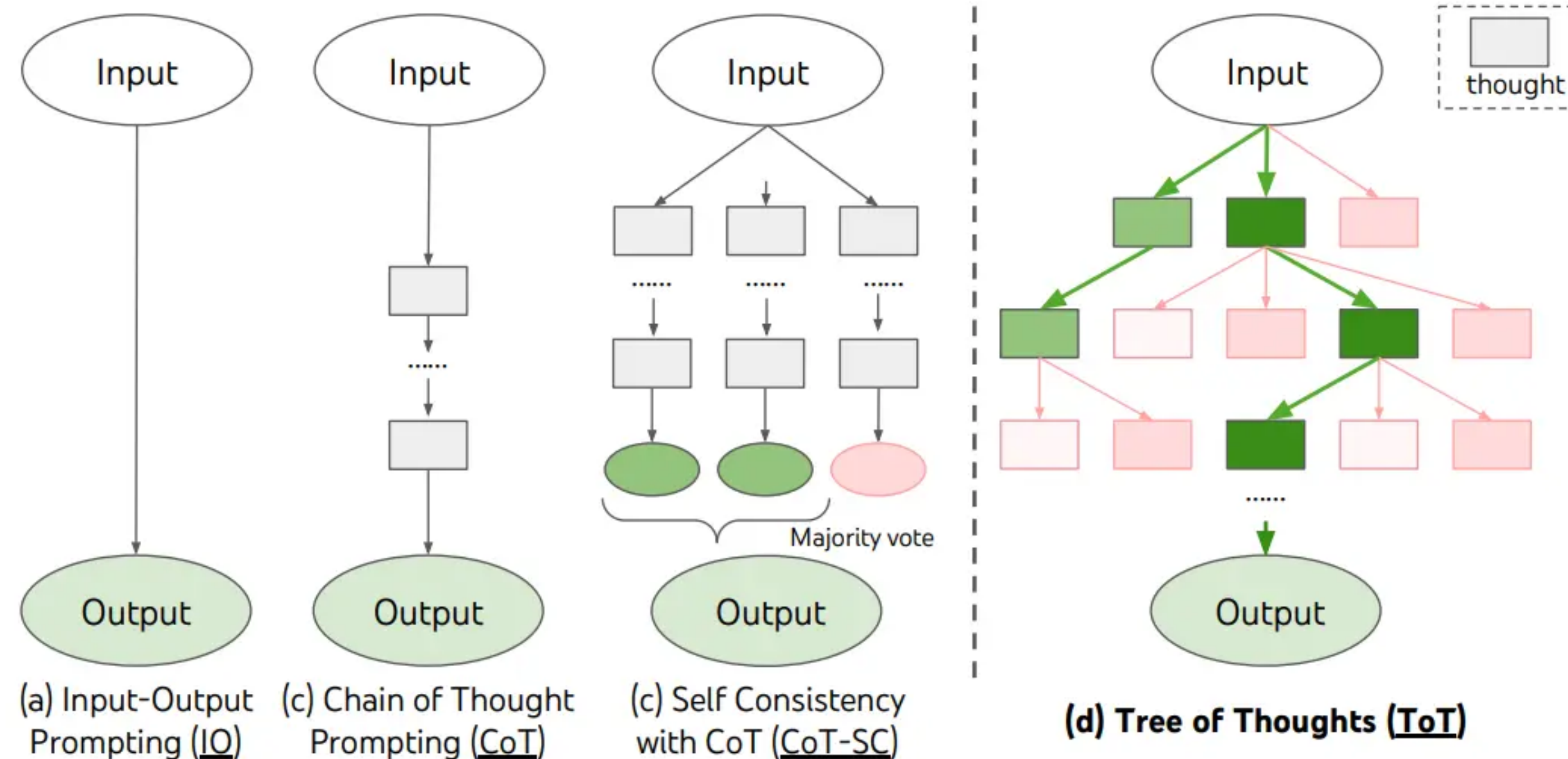
```
####
```

Output

```
<quotes>
- Chain-of-thought (CoT) prompting[27]
- Generated knowledge prompting[37]
- Least-to-most prompting[38]
- Self-consistency decoding[39]
- Complexity-based prompting[41]
- Self-refine[42]
- Tree-of-thought prompting[43]
- Maieutic prompting[45]
- Directional-stimulus prompting[46]
- Textual inversion and embeddings[59]
- Using gradient descent to search for prompts[61][62][63][64]
- Prompt injection[65][66][67]
</quotes>
```


Tree of Thoughts (ToT)

Tree-of-Thoughts = “brainstorm many → judge quickly → pursue the best”



Imagine three different experts are answering this question.
All experts will write down 1 step of their thinking,
then share it with the group.
Then all experts will go on to the next step, etc.
If any expert realises they're wrong at any point then they leave.
The question is...

Role/Persona

ReAct

Reason + Act

- ReAct reduces hallucination (it can look things up), provides step-level transparency, and works well for interactive tasks such as web search, code-execution, or game-playing

```
You are an agent that can think and act.
```

```
Available actions:
```

- `Search(query)` – look up information on the web
- `Calc(expr)` – evaluate a Python expression
- `Finish(answer)` – return your final answer

```
==== EXAMPLES ====
```

```
Q: Who wrote the novel "Dune"?
```

```
Thought: I should search the web.
```

```
Action: Search("Dune novel author")
```

```
Observation: "Dune" was written by Frank Herbert.
```

```
Thought: I have the answer.
```

```
Action: Finish("Frank Herbert")
```

Thanks